PCViT: A Pyramid Convolutional Vision Transformer Detector for Object Detection in Remote-Sensing Imagery

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***Abstract*— Remote-sensing object detection (RSOD) is a fundamental and valuable task in Earth monitoring. However, remote-sensing images (RSIs) are typically acquired from a bird's eye perspective, resulting in intrinsic properties such as complex backgrounds, random and dense distribution of objects, and multiscale objects. These properties hinder the direct application of well-performed detection methods in the natural images (NIs) domain to the RSIs domain, thereby limiting the attainment of desired performance. To address this, we propose a pyramid convolutional vision transformer (PCViT) that gets rid of the limitations of existing transformer methods. First, we employ a pyramid architecture to effectively capture the multiscale information present in RSIs. To enhance the feature extraction capabilities of the transformer, we introduce a parallel convolution module (PCM) that complements the local information that may be missed by the transformer. Furthermore, we propose a self-supervised pretraining strategy called multiperspective pretraining (MPP) to pretrain the model and subsequently finetune it on the downstream detection task. During the finetuning stage, we introduce a local/global *k*-NN attention (LGKA) to improve the token relationship establishment. In the neck part, we propose a feature-reflowing pyramid network (FRPN) to facilitate contextual information interaction and further enhance our PCViT's ability to process multiscale information. Experimental results on two representative datasets, namely NWPU VHR-10 and DIOR, demonstrate the effectiveness of our PCViT, as it achieves outstanding performance. These results highlight the suitability of PCViT for RSOD tasks.**

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***Index Terms*— Convolutional neural network (CNN), feature pyramid network (FPN), multiscale object detection, remote-sensing images (RSIs), vision transformer (ViT).**

I. Introduction

OBJECT detection remains a salient area of research within the remote-sensing (RS) field, characterized by its dual-focus on identifying the exact location of objects and categorizing them appropriately, offering significant practical value and broad application prospects, such as military identification [1], [2], [3], urban construction [4], [5], [6], and environment monitoring [7], [8], [9]. As show in Fig. 1, Contrary to natural images (NIs) observed horizontally, remote-sensing images (RSIs) are often captured from a bird's-eye view. This unique viewpoint gives rise to several distinct properties, including complex backgrounds, substantial variations in object sizes, and the difficulty in detecting small objects in dense scenes. Such inherent attributes of RSIs represent significant challenges to remote-sensing object detection (RSOD), calling for innovative methodologies to overcome them.

Addressing the challenges previously mentioned necessitates an accurate characterization of RSIs' attributes, alongside effective refinement of these features for downstream detection tasks. From universal detectors, such as R-CNN series [10], [11], SSD [12], and YOLO series [13], [14], [15], to dedicated detectors [16], [17], [18], [19] developed for RS scenes, the design of convolutional neural network (CNN)-based detectors is more and more in line with the requirements of RSOD tasks. For instance, Guo et al. [20] propose a unified multiscale CNN to handle objects of various scales in RSIs effectively. FMSSD [21] designs a spatial pyramid with several parallel dilated convolutional layers to enlarge receptive fields, which involves a large amount of calculation. Both methods make targeted improvements on the multiscale characteristics of RSIs, resulting in excellent performance, but lack consideration of the context information of the targets. In response, Li et al. [22] introduced a local-contextual feature fusion module to create robust joint representations conducive to the object detection task. Similarly, CAD-Net [16] implements a global context network alongside a pyramid local context network, with the aim of learning the global and local contexts of objects, respectively. Other models like GLNet [23] and FSoD-Net [24] also develop context modules to manage intricate scene issues and enhance detection performance.

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Fig. 1. Comparison between (a) NIs and (b) RSIs. Here, we show four common categories including vehicles, planes, ships, and bridges in two types of images. RSIs favor an overhead view compared to NIs viewed horizontally. The NIs are from the ImageNet-1K dataset, and the RSIs are selected from the NWPU VHR-10 datasets.

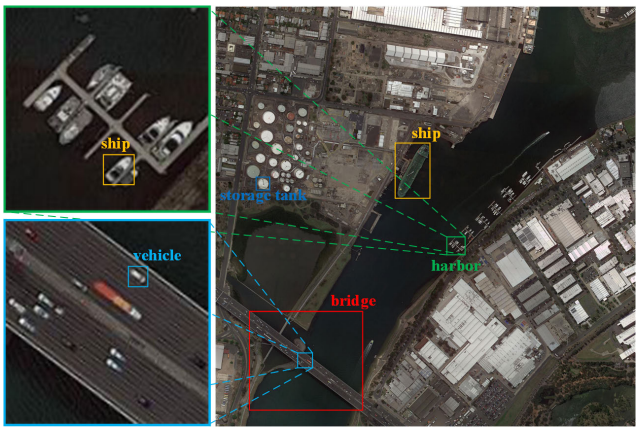


Fig. 2. Sample images from the DIOR dataset show that multiple objects (storage tanks, ships, bridges, harbors, and vehicles) in a single image are correlated.

Despite advances in design, the limitations of CNNs become apparent when considering the multiscale nature of RSIs and the context information of the targets. This limitation becomes more evident when detecting objects that occupy a large portion of the image such as bridges and harbors illustrated in Fig. 2, since these rely substantially on global information for detection. Therefore, there is an exigent demand for a more potent feature representation framework capable of handling such scenarios.

Recognized for its superior capabilities in modeling long-distance features, the transformer architecture offers a compelling solution to the dilemma faced by CNN-based detectors. The evolution of transformer-based detectors, such as ViTDet [25], DETR [26], Swin Transformer [27], and so on, including those [28], [29], [30], [31] specifically designed for RS scenes, aligns progressively with the demands of RSOD tasks. These detectors significantly improve the accuracy of detection, particularly for object detection within complex environments and large-scale objects. For instance, T-TRD [32] employs a CNN to extract local features and then utilizes a modified transformer to incorporate global information. GANsformer [33] adds a transformer branch to the CNN backbone to supply global features. Both methods recognize the requirement of both local and global information in RS tasks, but they merge local and global features only once, either sequentially or in parallel. This strategy may fail to capture deep underlying context dependencies fully, leading to subpar object detection in complex scenes. Furthermore, approaches like TransConvNet [31] and local perception Swin transformer (LPSW) [28] alternate between several transformer blocks and convolution blocks, fostering deeper feature interactions. Zheng et al. [34] propose a dynamically adaptive, one-stage detector based on a feature pyramid transformer, which embeds a transformer within the feature pyramid network (FPN) to enhance its feature fusion capacity. Although the above research attempts to fuse local and global features deeply to mitigate this issue, the separate processing of local and global information means that subsequent self-attention layers may overwrite local features. Additionally, the training of transformer-based networks tends to be challenging due to the lack of inductive bias, resulting in slower convergence.

Aimed at alleviating these bottlenecks, we propose a pyramid convolutional vision transformer (PCViT) detector as a novel solution in this article. Drawing inspiration from the hierarchical architecture of CNNs, our backbone consists of three scale stages. The initial two stages consist of convolutional blocks, which are adept at extracting shallow features and local information, and the final stage consists of transformer blocks, which are responsible for global modeling based on the local information extracted in the first two stages. At the same time, we also introduce a parallel convolution module (PCM) in each transformer block to make up for the missing local information in the transformer block. By using both serial and parallel connections, we cointegrate convolution and transformer components to deeply integrate local and global information and fully capture deep underlying context dependencies, thus achieving a comprehensive and detailed feature extractor.

Nevertheless, training deep-level and large models like vision transformers (ViTs) directly on downstream detection datasets is still challenging to achieve appropriate network weights. In response, several outstanding self-supervised methods [35], [36], [37], [38] have emerged, leveraging masked image modeling (MIM) on massive datasets to unleash the semantic understanding capabilities of ViTs. These methods assist in initializing ViTs and similar deep-level networks with improved weights. However, due to the hierarchical design of PCViT, traditional masking strategies are not applicable. To address this defect, we dedicate an MIM method named multiperspective pretraining (MPP). Additionally, considering the domain shift between RSIs and NIs, we initiate the pretraining on a large-scale RS dataset called MillionAID [39], followed by finetuning on downstream detection tasks. Moreover, our pretrained model can be effectively applied to other downstream tasks in RS, such as image classification and semantic segmentation, thus providing valuable contributions to the field of RS.

Additionally, in the top-down pathway of the FPN, the fusion between feature levels is limited to a single adjacent higher level, resulting in a tradeoff between semantic richness and detailed information. This limitation can lead to suboptimal results. To address this, we propose a novel approach called the feature-reflowing pyramid network (FRPN), which facilitates the interaction of information across different scales by leveraging up–down sampling and fusion operations. This enhancement proves beneficial in effectively guiding

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contextual information and significantly improving the detection performance of targets at various scales.

This study aims to bridge an existing gap in the field and offer valuable insights into the development of large-scale general ViTs specifically for the RS community. The main contributions of this article are specified as follows.

1) In general, a new powerful convolution transformer detector is proposed, which inherits the global information capture capability of the transformer, and the PCM supplements the local information missing from the transformer, which is extremely suitable for the multiscale objects of the RS scene. At the same time, a local/global k-NN attention (LGKA) is proposed to optimize the original self-attention, improving the representation ability and computational efficiency of multihead self-attention (MHSA).
2) To enhance the representation capability of our PCViT, a dedicated self-supervised pretraining approach called MPP is devised that caters to the hierarchical backbone structure. This method entails training the model on the extensive MillionAID dataset, effectively addressing domain disparities that arise when employing pretrained models trained on NIs in the field of RS. Furthermore, our pretrained model exhibits versatility and can be utilized for other RS downstream tasks like semantic segmentation and scene classification.
3) To tackle challenges related to the large-scale variations and difficulties in detecting small objects in RSIs, we devise an FPRN, which effectively leverages upsampling and downsampling operations to optimize the flow of information between different scale feature maps, thereby enhancing the network's ability to represent features of diverse object scales.

The rest of our article is structured as follows. Section II reviews related work on object detection. Section III details our proposed object detection framework. The experimental results of the proposed method are given in Section IV. Finally, Section V concludes this article.

II. Related Work

A. Object Detection in RSI

Object detection in RSI presents complexities due to environmental factors like weather, lighting, and clouds, impacting accurate detection. Challenges include occlusion, multiscale and multidirectional objects, and densely arranged objects. Despite this, the academic community has developed targeted solutions.

For instance, Li et al. [22] introduce multiangle anchors into the RPN network based on Faster R-CNN to handle rotation variations. They propose a dual-channel feature fusion network to extract local and contextual information, effective for addressing rotation variations. Addressing translation invariance in classification and translation variance in object detection, Zhong et al. [40] introduce a position-sensitive balancing framework using a fully convolutional network (FCN).

To address the detection of multiscale objects, especially small ones in high-resolution RSI, Guo et al. [20] propose a unified multiscale CNN. Additionally, FMSSD [21] adopts a spatial pyramid with parallel dilated convolutional layers to enlarge receptive fields, while CAD-Net [16] leverages spatial-and-scale-aware attention for feature pyramid construction across different scales. For densely packed object detection in RSIs, SCRDet [17] introduces a supervised multidimensional attention network to mitigate background noise's adverse effects. ClusDet [41] introduces a cluster proposal subnet for predicting cluster regions through a supervised process.

Although these deep-learning models effectively tackle object detection, their feature characterization capabilities have limitations. Addressing this, the transformer architecture is a promising alternative to overcome these constraints.

B. Vision Transformer

The transformer architecture emerges as an outcome of a thorough exploration of the attention mechanism. Initially incorporated in early CNN networks to enhance region-specific attention, the attention mechanism's applicability is extended to natural language processing (NLP) due to its potent global representation capability. Recently, the transformer architecture, specifically ViT [42], has been introduced to computer vision for tasks like object detection.

ViT partitions a 2-D image into fixed-size patches, each containing a position code for token generation. These tokens are then fed into the standard transformer encoder, consisting of multiple MHSA and MLP blocks. These components enhance crucial features within each patch while capturing its broader relationships with other patches. Unlike the CNN and generative adversarial network (GAN), the transformer excels at capturing middle-range and long-range dependencies without relying on past hidden states or recurrence, which is essential in RSOD. Given the extensive variability in ground backgrounds and target distributions in RSIs, robust global dependency modeling is crucial for effective detection.

While ViT exhibits promising performance in image classification, its single feature layer output structure is not ideal for dense prediction tasks like object detection and semantic segmentation. In response, several ViT variants are introduced to enhance its applicability to dense prediction tasks. For instance, PvT [43] introduces a pyramid structure into the Transformer, establishing a dedicated backbone for dense prediction tasks. Swin Transformer [27] introduces a hierarchical transformer with shifted windows for representation computation. ViTDet [25], retaining ViT's original architecture, achieves multiscale output through up- and downsampling in the final feature layer, thereby maximizing the potential of pretraining.

C. Masked Image Modeling

The field of MIM gains prominence with the introduction of the masked autoencoders (MAE) architecture [44], [45], mirroring the masked language modeling (MLM) task in NLP. Earlier MIM research can be broadly categorized into two domains: regression targets and masking strategy.

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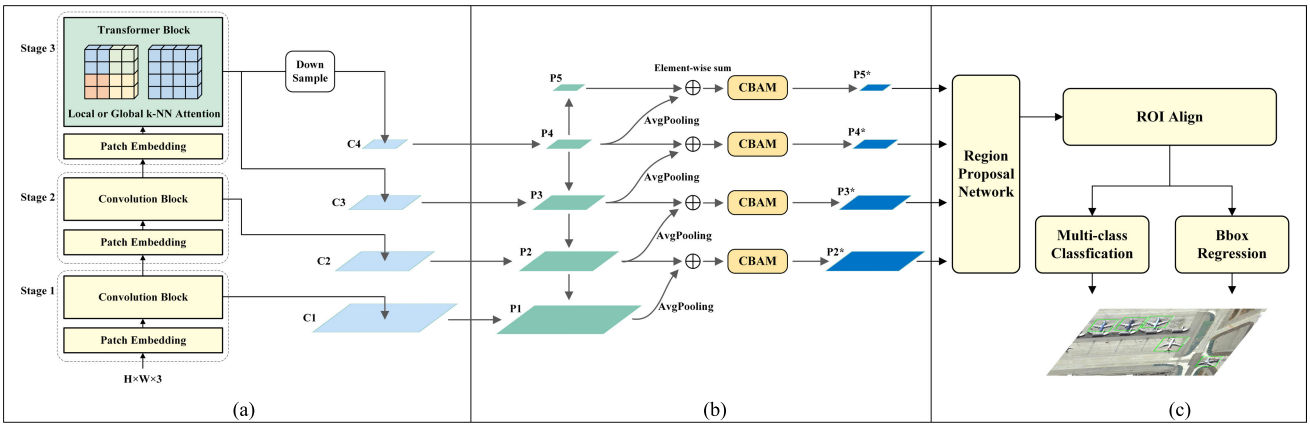


Fig. 3. Structure of the baseline of the proposed PCViT. The proposed backbone constitutes a multiscale pyramid with three scale stages. The initial two stages consist of convolutional blocks, and the final stage consists of transformer blocks. Here, we refine the transformer block using the PCM and LGKA modules. Then, the multiscale features derived from the backbone are fed into the subsequent FRPN neck to facilitate contextual information interaction before being directed to the detection head. (a) Backbone with PCViT. (b) Feature-reflowing pyramid. (c) Faster R-CNN detection head.

In terms of regression targets, MAE [35] and SimMIM [36] directly employ RGB values of original pixels as regression targets. BEiT [46], conversely, utilizes a trained deep variational autoencoder (dVAE) network, creating a challenging task by predicting visual tokens of masked image patches. Maskfeat [37] explores various feature forms and finds that HOG features applied to masked tokens yield notable performance and efficiency gains.

Concerning masking strategy, SiT [47], MC-SSL0.0 [48], and BEiT [46] adapt a block-wise masking approach, masking a spatially arranged block of neighboring tokens. MAE [35] and SimMIM [36], conversely, utilize random masking with large patch sizes or a substantial proportion of masked patches. SemMAE [38] goes further by explicitly learning semantic parts to create meaningful hints for MIM.

The employed masking strategies above are not readily applicable to multiscale ViTs, due to the necessity of retaining all tokens in subsequent conversion stages. This results in an impractical computational burden for pretraining substantial and intricate models, negating the efficiency benefit of MAE in excluding masked tokens from transformer encoders. Moreover, direct pretraining using a convolutional transformer encoder introduces disparities in pretraining–finetuning, as only visible tokens are processed during the finetuning stage.

III. Methodology

This section introduces the PCViT network, outlining its architecture and design specifics. To optimally harness the network's potential, a novel MPP framework tailored for PCViT is elaborated upon, detailing the masking strategy and pretraining configurations. Additionally, it explores the finetuning process of the pretrained model for the RSOD task, leveraging the LGKA module to enhance MHSA's performance and efficiency. Concurrently, it delineates the FRPN module, aimed at bolstering the network's detection capabilities across diverse object scales within the PCViT framework, effectively addressing challenges posed by scale variations.

A. PCViT

1) Pyramid Convolutional Vision Transformer: As shown in Fig. 3(a), we designed a hybrid convolutional transformer network with a simple but efficient architecture, which consists of three stages with output spatial resolutions of $(H/4) \times (W/4)$, $(H/8) \times (W/8)$, and $(H/16) \times (W/16)$, respectively. Here, $H \times W$ signifies the input image resolution. In each stage, tokens are downscaled through a stride-2 convolution, halving their preceding spatial resolution. Reflecting the convolution's efficacy in capturing shallow features and the transformer's aptitude for global modeling in deeper features, we employ efficient convolutional blocks in the initial two stages and potent transformer blocks in the final stage. Our convolutional block is designed in alignment with the transformer block's principles, with the only alteration being the substitution of the self-attention mechanism with a 5×5 depthwise convolution. Diverging from the standard ViT's MHSA block, our transformer block incorporates a PCM to augment the local information that the transformer lacks. Following this three-stage process, token embeddings are obtained as $E_1 \in \mathbb{R}^{(H/4)\times(W/4)\times C_1}$, $E_2 \in \mathbb{R}^{(H/8)\times(W/8)\times C_2}$ and $E_3 \in \mathbb{R}^{(H/16)\times(W/16)\times C_3}$, respectively.

2) Transformer Block With PCM: The general transformer block has a strong global modeling ability and can collect global information, but lacks local information. To this end, we design a PCM to complement the missing local information in the transformer block. Thereby, our transformer block can well model long-distance dependencies and take into account the local information of features at the same time.

We denote that the input feature of the ith transformer block is $f_i \in \mathbb{R}^{(H/16)\times(W/16)\times C_3}$. f_i passes through two branches, one is MHSA, which models the long-distance global information in f_i. Specifically, the input tokens f_i are first projected to queries (Q), keys (K), and values (V) through the projection matrix, that is, $Q, K, V = f_i W_Q, f_i W_K, f_i W_V$, where $W_{Q/K/V} \in \mathbb{R}^{C_3\times C_3}$ refers to the projection matrix of Q, K, V, respectively, and then the self-attention operation is calculated as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{D}}\right)V. \tag{1}$$

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The attention module is iterated h times to construct the MHSA module, with h representing the number of heads. The resultant features from the h heads are concatenated along the channel dimension, forming the MHSA module's output. Upon passage through MHSA, the global information of f_i is obtained, denoted as follows:

$$f_i^g = \mathrm{MHSA}(f_i). \tag{2}$$

The other is a PCM, responsible for capturing local information within f_i. As self-attention operates on 1-D sequences and convolution operates on 2-D images, we initially transform the sequence into its corresponding image format. Subsequently, convolution is applied to extract spatial local information, followed by reconversion of the convolution result into a 1-D sequence

$$f_i^l = \mathrm{PCM}_i(f_i) = \mathrm{Img2Seq}(\mathrm{Conv}_i(\mathrm{Seq2Img}(f_i))) \tag{3}$$

where $\mathrm{PCM}_i(\cdot)$ represents the PCM, composed of a Seq2Img$(\cdot)$ operation, three successive convolutional layers, and an Img2Seq$(\cdot)$ operation. This configuration imbues token features with the local context.

The integration of global and local information is achieved by straightforwardly adding the global feature post-MHSA and the local feature post-PCM

$$f_i^{gl} = f_g + f_l. \tag{4}$$

After the fused features pass through the FNN, they are input to the next transformer block

$$f_{i+1} = \mathrm{FFN}\left(f_i^{gl}\right) + f_i^{gl}. \tag{5}$$

B. Self-Supervised Pretraining by MPP

Given the multiscale nature of objects in RSIs, a pyramid backbone is devised, rendering the conventional random masking strategy unfeasible. To address this, we employ block-wise masking with masked convolutions and introduce a multiperspective masking (MPM) strategy.

1) Block-Wise Masking With Masked Convolutions: Applying a uniform mask to input tokens of stage 1 from the $(H/4) \times (W/4)$ feature map leads to partial visibility of information in all stage 3 tokens, necessitating the retention of all stage 3 tokens. However, this would result in substantial decoder computation, contradicting the MAE principle. Thus, we propose generating initial random masks to cover 75% of stage 3 input tokens. These masks are then upsampled by factors of 2 and 4 to yield corresponding block-wise masks for stages 2 and 1 inputs. During encoding, the associated masked tokens across the three stages are omitted, and their reconstruction is performed by the decoder to facilitate feature learning. Consequently, MPP retains only a minimal 25% of tokens in the resource-intensive transformer block for training, thereby maintaining MPP's efficiency intact.

Nonetheless, the inherent 5×5 depth-wise convolutions in the initial two stages inherently yield a receptive field larger than the masked patch, which can result in information leakage during masked token reconstruction. To mitigate this risk and ensure the quality of pretraining, we employ masked convolutions in the first two stages. This approach guarantees that masked regions remain uninvolved in the encoding process. The utilization of masked convolutions is pivotal for the remarkable performance of MPP, preventing any discrepancies between pretraining and finetuning scenarios by excluding partially masked tokens from the stage.

2) Multiperspective Masking: The objective of MPM is to sequentially extract several masked perspectives based on predefined rules, thereby augmenting the overall proportion of tokens utilized for reconstruction. As augmented perspectives decrease the samples within a designated batch size, we adapt the learning rate scaling strategy to enhance model training for improved generalization.

MPM Regulation: Suppose $\mathcal{M}_i^1 = \mathbb{M}(\mathcal{X}_i, r_{\mathrm{mask}})$, where r_{mask} is the masking rate. The set of tokens masked in the preceding $K-1$ masking is defined as $\mathcal{M}_i^{K-1} = 1 - \prod_{k=1}^{K-1}(1 - M_i^k)$, where $K \geq 2$. Therefore, MPM$(\mathbb{M}_{\mathrm{mpm}})$ with the MPM regulation can be formulated as

$$\begin{aligned} &\mathbb{M}_{\mathrm{mpm}}\left(\mathcal{X}_i, r_{\mathrm{mask}}, \mathcal{M}_i^{K-1} = M_i^K\right) \\ &\text{s.t. } M_i^K \cdot \left(1 - \mathcal{M}_i^{K-1}\right) \neq 0. \end{aligned} \tag{6}$$

Equation (6) can be established exclusively when M_i^K can mask tokens that have not been masked before. To sample the Kth mask based on this rule, the image tokens are initially divided into two distinct subsets: $\mathcal{X}_i \cdot \mathcal{M}_i^{K-1}$ (previously masked tokens) and $\mathcal{X}_i \cdot (1 - \mathcal{M}_i^{K-1})$ (currently unmasked tokens), as shown in Fig. 4.

Subsequently, tokens are partially sampled for masking from the unmasked token set using a positive masking rate $r_{\mathrm{mask}} > 0$, while the remaining tokens are sampled from the masked set. This approach ensures that each masked perspective includes novel tokens beyond previous perspectives, thus maintaining a fixed masking rate and simultaneously enhancing the overall prediction rate for each image.

3) Pretraining on MillionAID [39]: The MillionAID dataset is a new large-scale benchmark dataset containing a million instances for RS image scene classification. Diversity, richness, and scalability (DiRS) could be considered the basic principles when creating benchmark datasets for RS image interpretation. MillionAID contains a wide range of semantic categories, that is, 51 scene categories. The number of images in each scene category ranges from about 2000 to 45 000. It contains 1 000 848 nonoverlapping RS scenes in RGB formats, which is suitable to serve as inputs for typical deep vision models. MillionAID is collected from Google Earth, where images are captured with various sensors and thus have different resolutions. Generally, the image sizes in the MillionAID dataset range from 100×100 to $31\,672 \times 31\,672$ pixels. It should be noted that although the MillionAID dataset contains both images and labels, we only use the images during pretraining and discard the labels by following the unsupervised pretraining routine.

We utilized all images from the MillionAID test set, totaling 990 848 images, for self-supervised pretraining. Our pretraining phase operated with the AdamW [49] optimizer, employing a batch size of 128, distributed across 4 NVIDIA GeForce RTX 3090 GPUs. In line with the default settings established in MAE, our model underwent 100 epochs during

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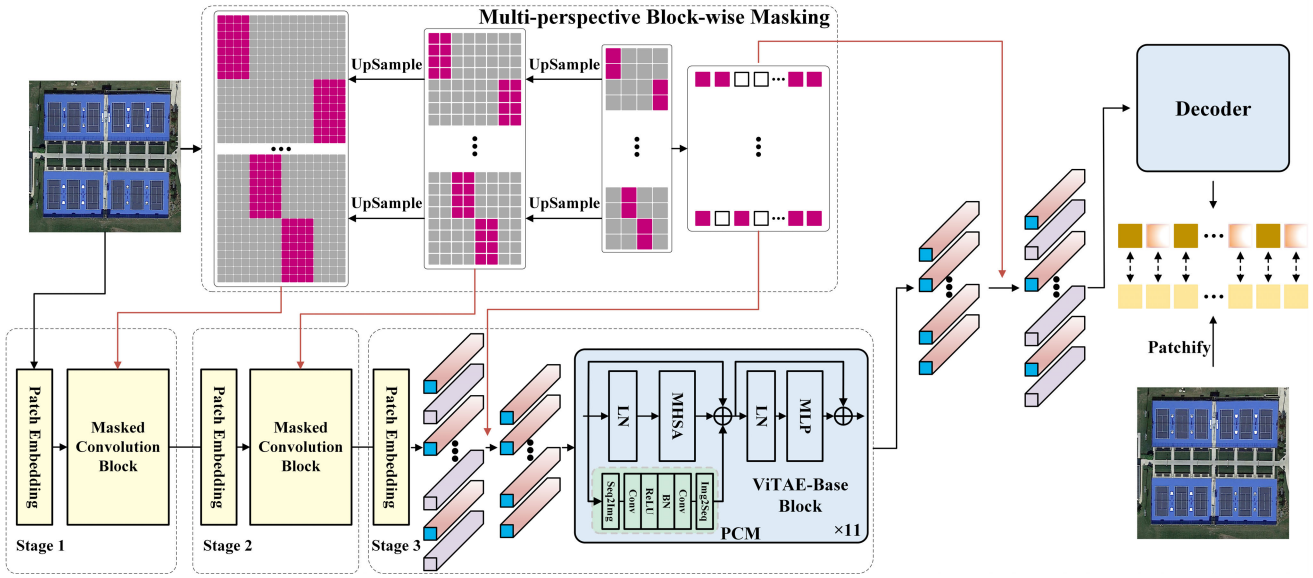


Fig. 4. Pipeline of the proposed MPP. During pretraining, K masked perspectives of each image are randomly sampled in a mini-batch with MPM. Then, they will be fed to the encoder and the decoder for invisible reconstruction with targets.

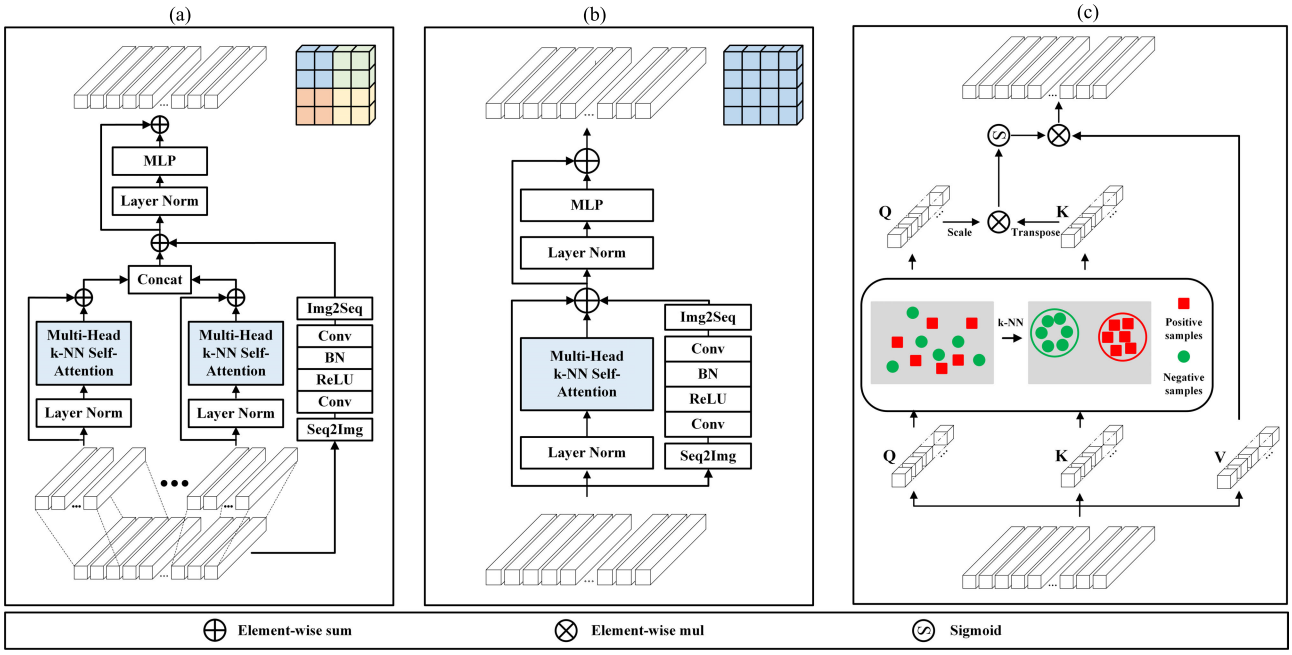


Fig. 5. LGKA. In each group of transformer subblocks, we use local attention for the first two layers, that is, reduce computational complexity through 16 × 16 window attention. For propagation between windows, we use global attention in the third layer. (a) Local k-NN attention. (b) Global k-NN attention. (c) Multihead k-NN self-attention.

training unless otherwise specified. Considering computational resource and training duration constraints, the mask ratio set at 0.75 for pretraining followed the empirical findings of MAE [35].

Finetuning was conducted by adjusting parameters from the backbone network and classifier using a layer-wise learning rate decay mechanism [46]. Our finetuning settings were mostly consistent with MAE [35], except for variations in batch size and epochs. The evaluation of model performance was directly carried out within the RS target detection task, specifically employing the DIOR dataset for finetuning the pretrained model.

The experimental outcomes revealed the efficacy of adopting the MPP architecture. Our PCViT's pretrained model showcased superior performance during downstream task finetuning settings, exhibiting a significant 2.17% improvement in mAP_{50} compared to the pretrained ViT-B model.

C. Finetuning With LGKA

1) k-NN Attention: The complete term for k-NN attention is k-nearest neighbor attention. In Fig. 5(c), instead of computing the attention matrix for all the query–key pairs as in vanilla attention, we select the top-k most similar keys and values for each query in the k-NN attention. There are two versions of

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k-NN attention, as described below. In contrast to computing the attention matrix for all query–key pairs as in standard attention, the k-NN attention selects the top-k most similar keys and values for each query. There are two versions of k-NN attention, as outlined below.

Slow version: For the ith query, the Euclidean distance is computed against all keys, resulting in the retrieval of its k-nearest neighbors $\mathcal{N}_i^K$ and $\mathcal{N}_i^V$ from keys and values, respectively. Subsequently, the scaled dot product attention is calculated as follows:

$$\mathbf{A}_i = \text{softmax}\left(\frac{< \mathbf{q}_i, \left(\mathbf{k}_{j1}, \ldots, \mathbf{k}_{jl}, \ldots, \mathbf{k}_{jk}\right) >}{\sqrt{d}}\right) \tag{7}$$

where $\mathbf{k}_{jl} \in \mathcal{N}_i^k$. The shape of the final attention matrix is $\mathbf{A}^{knn} \in \mathbb{R}^{n\times k}$, and the new values $\hat{\mathbf{V}}^{knn}$ are of the same size of values $\hat{\mathbf{V}}$. The "slow version" provides the precise definition of k-NN attention. However, it is notably slow as it requires distance computations for different k keys for each individual query.

Fast version: To address the computational inefficiency of calculating Euclidean distances for all keys per query, we introduce a more efficient variant of k-NN attention. This approach leverages matrix multiplication operations. Similar to vanilla attention, the dot product is employed to compute all queries and keys. Subsequently, the top-k elements are chosen row-wise for softmax computation. This process can be formulated as

$$\mathbf{A}_i = \text{softmax}\left(\tau_k\left(\frac{\mathbf{QK}^T}{\sqrt{d}}\right)\cdot \mathbf{V}\right) \tag{8}$$

where $\tau_k(\cdot)$ denotes the row-wise top-k selection operator.

2) Local/Global Attention: Local attention: In Fig. 5(a), given an input $X \in \mathbb{R}^{C\times H\times W}$, local attention first partitions it into several nonoverlapping windows, that is, $X_w \in \mathbb{R}^{C\times s\times s}$, where s represents the window sizes. There are a total of $(H/s)\times(W/s)$ windows in the image. Then, features inside each window are transformed to h triplets representing the query, key, and value features: $\mathbf{Q}_w^{(i)}, \mathbf{K}_w^{(i)}, \mathbf{V}_{w\,i=1}^{(i)h}$ by three linear layers, respectively, where $\mathbf{Q}_w^{(i)}, \mathbf{K}_w^{(i)}, \mathbf{V}_w^{(i)} \in \mathbb{R}^{C'\times s\times s}$, $C = hC'$, and h is the number of heads in the attention layers. We denote the generated features as $\mathbf{Q}_w^{(i,j)}, \mathbf{K}_w^{(i,j)}, \mathbf{V}_w^{(i,j)} | j = 1, \ldots, (HW/s^2)_{i=1}^h$, where j indices the window. After that, the attention calculations are conducted inside each window, that is,

$$\begin{aligned}\mathbf{F}_w^{(i,j)} &= \text{Attention}\left(\mathbf{Q}_w^{(i,j)}, \mathbf{K}_w^{(i,j)}, \mathbf{V}_w^{(i,j)}\right)\\ &= \text{softmax}\left(\frac{\mathbf{Q}_w^{(i,j)}\mathbf{K}_w^{(i,j)^T}}{\sqrt{d}}\right)\mathbf{V}_w^{(i,j)'}\end{aligned} \tag{9}$$

where $\mathbf{F}_i^{(i,j)} = (R)^{sw}\times C'$. Then, the features from different heads are concatenated along the channel dimension, and features from different windows are concatenated along the spatial dimension to recover the shape of the input feature.

Global attention: As shown in Fig. 5(b), the global attention is exactly the same as the transformer block of ViT, which is used for the first, fourth, seventh, and tenth layers of the transformer block for information exchange between local attention.

D. Feature-Reflowing Pyramid Network

We introduce an innovative FRPN that distinguishes itself from FPN, resulting in enriched feature representation and improved outcomes. Beyond the comprehensive semantic features extracted from higher levels, this novel pyramid structure acquires detailed spatial information via a straightforward yet effective feature-reflowing pathway. Leveraging its potent expressive capacity, the feature-reflowing pyramid structure significantly enhances the network's performance in multiclass and multiscale object detection.

We now delve into the central component, the feature-reflowing pyramid structure, depicted in Fig. 3(b). To capture finer details for small objects, we introduce the fusion of the two lower levels $[\mathrm{P}_1, \mathrm{P}_2]$ into the top-down pathway, in contrast to the original FPN, which conducts fusion solely based on $[\mathrm{P}_2, \mathrm{P}_3, \mathrm{P}_4, \mathrm{P}_5]$. In the top-down pathway, the fundamental feature maps $[\mathrm{P}_1, \mathrm{P}_2, \mathrm{P}_3, \mathrm{P}_4, \mathrm{P}_5]$ generated by FPN already encompass substantial semantic information. Building upon these levels, the feature-reflowing pathway is established for each scale. Initially, average-pooling is applied to reduce the dimensions of $[\mathrm{P}_2, \mathrm{P}_3, \mathrm{P}_4, \mathrm{P}_5]$ and obtain intermediate feature maps identical in size to $[\mathrm{P}_3, \mathrm{P}_4, \mathrm{P}_5]$, except for P_1 which shares the same spatial dimensions as P_2. Subsequently, the element-wise sum is used to fuse $[\mathrm{P}'_2, \mathrm{P}'_3, \mathrm{P}'_4, \mathrm{P}'_5]$ with the intermediate feature maps, enabling the incorporation of comprehensive information from their lower counterparts.

To streamline information and enhance feature quality, the convolutional block attention module (CBAM) is employed to refine fused features across channel and spatial dimensions, yielding output feature maps $[\mathrm{P}^*_1, \mathrm{P}^*_2, \mathrm{P}^*_3, \mathrm{P}^*_4]$. Specifically, P^*_3, exemplified in Fig. 3, serves as a representation for elucidating the processing mechanism within the feature-reflowing pathway. This CBAM module comprises two sequential submodules: channel attention (C_a) and spatial attention (S_a). Their computations are separate as follows:

$$C_a = \sigma\left(\text{MLP}(\text{AvgPool}(\mathrm{P})) + \text{MLP}(\text{MaxPool}(\mathrm{P}))\right) \tag{10}$$

$$S_a = \sigma\left(W_{7\times7}\left(\left[\text{AvgPool}(\mathrm{P}); \text{MaxPool}(\mathrm{P})\right]\right)\right). \tag{11}$$

Here, P signifies the input feature maps, with MLP representing the multilayer perceptron, $W_{7\times7}$ indicating a convolution operation utilizing a 7×7 convolution kernel, and σ denoting the sigmoid activation function.

This construction embodies the feature-reflowing structure in the following manner:

$$\mathrm{P}'_i = \begin{cases}\mathrm{P}_i + \mathrm{P}_{i-1}, & \text{if } i = 1;\\ \mathrm{P}_i + \text{AvgPool}(\mathrm{P}_{i-1}), & \text{if } i = 2, 3, 4.\end{cases} \tag{12}$$

$$\mathrm{P}^*_i = \mathrm{P}'_i \times C_a\left(\mathrm{P}'_i\right) \times S_a\left(\mathrm{P}'_i \times C_a\left(\mathrm{P}'_i\right)\right). \tag{13}$$

IV. Experimental Results

To demonstrate the effectiveness of our proposed PCViT network, we conducted experiments on two representative benchmarks, NWPU VHR-10 [50] and DIOR [51], and compared them with existing SOTA object detection algorithms, both of which showed the superiority of our proposed network. At the same time, a full ablation experiment was carried out to

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TABLE I
NWPU VHR-10 DATASET DETAILS

Class		Samples Number	
No.	Name	Train	Test
1	Airplane	634	123
2	Ship	231	71
3	Storage tank	509	146
4	Baseball diamond	320	70
5	Tennis court	429	95
6	Basketball court	126	33
7	Ground track field	136	27
8	Harbor	125	99
9	Bridge	93	31
10	Vehicle	543	55
Total		3,146	750

demonstrate the significance of each module of the network. We will describe the dataset, experimental setup, ablation experiments, and so on in detail.

A. Datasets

To progress the evolution of object detection in RSI, a variety of public datasets [52], [53], [54], [55] have emerged consecutively. To validate the efficiency and supremacy of our proposed network, we utilize the following two datasets, which are widely employed due to their diversity, to conduct experimental evaluations.

1) NWPU VHR-10 [50]: The NWPU VHR-10 dataset, initially presented by Cheng et al. [50] at Northwestern Polytechnical University, encompasses 800 high-resolution RSIs sourced from Google Earth and Vaihingen data. These images exhibit varying dimensions and spatial resolutions, ranging from 533 × 597 to 1728 × 1028 and spatial resolutions spanning 0.08–2 m. Among these 800 images, 650 positive samples are meticulously annotated with horizontal bounding boxes (HBB), totaling 3896 object instances, while 150 negative samples do not encompass any objects of interest. The division of the positive samples into a training set and testing set with an 8 : 2 ratio, consisting of 520 and 130 images, respectively. Detailed descriptions of dataset categories are shown in Table I.

2) DIOR [51]: Given the limited sample size of the NWPU VHR-10 dataset, we extended our evaluation to include the DIOR dataset, to ascertain the efficacy and generalizability of our proposed methodology. DIOR, an expansive dataset of aerial images annotated with HBB, encompasses objects with considerable variability in size. This variability extends not only to spatial resolutions, ranging from 0.5 to 30 m, but also to interclass and intraclass size disparities. DIOR's complexity is further underscored by diverse imaging conditions, weather patterns, seasonal changes, image quality variations, and other factors, contributing to pronounced interclass similarity and intraclass diversity. To be specific, the dataset comprises 23 463 images and 192 472 instances, spanning 20 distinct object categories, each containing approximately 1200 images. The training, validation, and test sets comprise 5862, 5863, and 11 738 images, respectively. Comprehensive details of dataset categories are provided in Table II.

TABLE II
DIOR DATASET DETAILS

Class		Samples Number	
No.	Name	Train	Test
1	Airplane	1,238	650
2	Airport	328	334
3	Baseball field	1,245	1,139
4	Basketball court	561	516
5	Bridge	506	861
6	Chimney	342	307
7	Dam	252	260
8	Expressway service area	537	543
9	Expressway toll station	292	318
10	Golf field	237	274
11	Ground track field	565	597
12	Harbor	1,147	1,217
13	Overpass	577	753
14	Ship	13,028	14,323
15	Stadium	293	302
16	Storage tank	1,337	1,705
17	Tennis court	2,233	2,665
18	Train station	246	255
19	Vehicle	6,709	7,016
20	Wind mill	961	1,404
Total		32,634	35,439

TABLE III
ABLATION EXPERIMENTS ABOUT PRETRAINING

No.	ViT-B [42]	Pretrained ViT-B	PCViT	Pretrained PCViT
1	90.83	**93.03**	84.30	92.04
2	49.64	83.54	24.48	**84.18**
3	91.81	92.75	87.16	**92.89**
4	65.43	83.03	48.95	**88.10**
5	26.82	51.02	19.43	**59.49**
6	86.42	**90.96**	73.53	90.67
7	45.16	69.85	16.44	**71.82**
8	63.83	68.91	44.68	**69.99**
9	42.42	58.89	29.64	**62.90**
10	65.93	82.90	19.38	**85.02**
11	62.64	80.56	41.47	**84.69**
12	38.20	57.50	29.47	**60.23**
13	48.40	66.61	42.37	**70.64**
14	73.67	79.36	67.73	**80.36**
15	92.18	95.23	82.28	**95.57**
16	76.20	83.87	69.42	**84.99**
17	83.24	89.83	78.05	**90.93**
18	34.22	67.89	21.32	**68.27**
19	61.07	74.55	47.41	**79.56**
20	66.38	**79.60**	16.21	78.94
mAP_{50}	63.22	77.49	47.19	**79.56**

B. Experimental Setup

Our experimental environment is the PyTorch framework (PyTorch1.9.0) in the Ubuntu 20.04 operating system, and all experiments are carried out on two NVIDIA GeForce RTX 3090 GPUs, each with 24 GB of memory. Our network uses PCViT pretrained on the MillionAID [39] large-scale remote-sensing dataset as the backbone. As for the backbone of other models, ViTAE-B [30] also uses the weights pretrained on MillionAID, and ResNet50 [56], ViT-B [42], Swin-B [27], and so on all use the weights pretrained on the ImageNet-1K classification task.

In terms of data augmentation, we uniformly resize the input image to 1024 × 1024, use 0.5 horizontal flips as data augmentation, and no other tricks. Following the de facto standard, all models are trained using the AdamW [49] optimizer with learning rate and weight decay set to 0.0001 and 0.1, respectively. Momentum is set to 0.9, representing the weight

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of the last updated gradient. To train more fully, we use a 6× training schedule with 72 epochs and a batch size of 1. The learning rate is adjusted by a multistep scheduler, that is, the learning rate is reduced by 10× at the 48th and 66th epochs. We also employ a hierarchical learning rate decay strategy, where the decay rate is set to 0.7.

C. Evaluation Metrics

The Precision–Recall curve (PRC) and average precision (AP), which are standard and authoritative evaluation indicators, are adopted to quantitatively evaluate different methods in RSOD.

1) Precision–Recall Curve: It is necessary to introduce the confusion matrix to evaluate different methods. The confusion matrix consists of four parts, which are true positive (TP), true negative (TN), false positive (FP), and false negative (FN).

Then, Precision and Recall are expressed as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{15}$$

Additionally, after checking the category, comparing locations is inevitable to distinguish whether the predicted box is a TP sample or an FP sample. Intersection-over-union (IoU) between the ground truth (GT) and the predicted result (PR) is a crucial measurement for comparing location, which can be described as

$$\text{IoU} = \frac{\text{GT} \cap \text{PR}}{\text{GT} \cup \text{PR}}. \tag{16}$$

To be specific, the predicted boxes with IoU bigger than 0.5 are sentenced to be positive and the rest are negative.

2) Average Precision: The widely used performance evaluation metrics for object detection are the AP of each class and the mean average precision (mAP) of all classes. Generally, AP is the integration of the PRC from 0 to 1 on recall, which is a more comprehensive and robust indicator of the detection method performance. For a detector, the mAP is higher, the detection performance is better obviously. The AP and mAP are defined as

$$\text{AP} = \int_0^1 \text{P(R)} d\text{R} \tag{17}$$

$$\text{mAP} = \frac{1}{N_{\text{cls}}} \sum_{i=1}^{N_{\text{cls}}} \text{AP}_i \tag{18}$$

respectively, where P and R refer to Precision and Recall, and N_{cls} represents the total number of classes. Defined in the MS COCO benchmark [57], mAP_{50} means the IoU threshold of defining TP is 0.5. Note that mAP_{50} takes objects of all scales into consideration. Moreover, mAP_s, mAP_m, and mAP_l are for small-, medium-, and large-scale evaluation in NWPU VHR-10 [50] and DIOR [51], respectively.

D. Experimental Results and Analysis

In this section, we first conducted ablation experiments on the necessity of deep network pretraining such as ViT-B and PCViT on DIOR. Then, using PCViT pretrained on the MillionAID dataset as the backbone, in the finetune stage, the performance of the two optimization modules proposed for PCViT, including the LGKA and the FRPN, is quantitatively and qualitatively evaluated. Finally, a comparison between our proposed method and the state-of-the-art methods on the NWPU VHR-10 dataset and the DIOR dataset is shown. The detection results on different datasets are shown in Tables V and VI. In the table, the data in bold indicates the maximum value, and the data in underline indicates the second maximum value.

TABLE IV
ABLATION EXPERIMENTS ON THE DIOR DATASET

No.	PCViT	+LGKA	+FRPN	+LGKA&FRPN
1	92.04	92.82	**92.90**	**92.90**
2	84.18	84.36	83.10	**84.49**
3	92.89	93.74	93.69	**93.77**
4	**88.10**	87.00	87.71	87.59
5	59.49	59.02	58.51	**59.63**
6	90.67	**92.74**	91.69	91.78
7	71.82	72.92	73.42	**74.47**
8	69.99	69.67	**71.31**	70.28
9	62.90	62.71	62.78	**62.92**
10	85.02	84.26	**86.44**	84.56
11	84.69	84.65	**84.86**	**84.86**
12	60.23	60.40	59.39	**61.40**
13	70.64	70.29	**70.79**	70.27
14	80.36	80.30	**80.51**	80.37
15	95.57	95.24	95.26	**96.17**
16	84.99	85.63	84.83	**85.74**
17	90.93	91.88	91.85	**92.77**
18	68.27	69.60	71.17	**73.35**
19	79.56	**79.60**	79.56	79.56
20	78.94	**80.03**	78.07	78.11
mAP_{50}	79.56	79.84	79.94	**80.25**
mAP_s	20.09	20.47	21.06	**21.78**
mAP_m	55.85	55.12	56.18	**58.24**
mAP_l	90.51	90.69	90.53	**90.96**

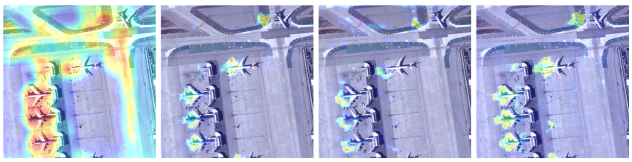


Fig. 6. Heatmaps of different versions of PCViT. (a) PCViT+FPN without pretraining. (b) PCViT+FPN with pretraining. (c) PCViT+LGKA+FPN with pretraining. (d) PCViT+LGKA+FRPN with pretraining.

1) Ablation Experiments: Initially, we establish the importance of deep network pretraining, focusing on two backbones: the classic ViT-B and our novel PCViT. Ablation experiments, conducted with consistent parameter settings on the DIOR dataset, reveal profound implications of pretrained weights on deep networks. ViT-B's performance on DIOR gains a 14.27% enhancement upon integrating pretrained weights, while PCViT showcases more pronounced progress, with a notable 31.63% increase in mAP_{50} on DIOR. These results underscore PCViT's complexity, with its intricate structure and deeper network layers demanding superior initial pretrained weights to facilitate convergence.

Moreover, when comparing ViT-B with our proposed PCViT, both leveraging pretrained weights, it becomes evident

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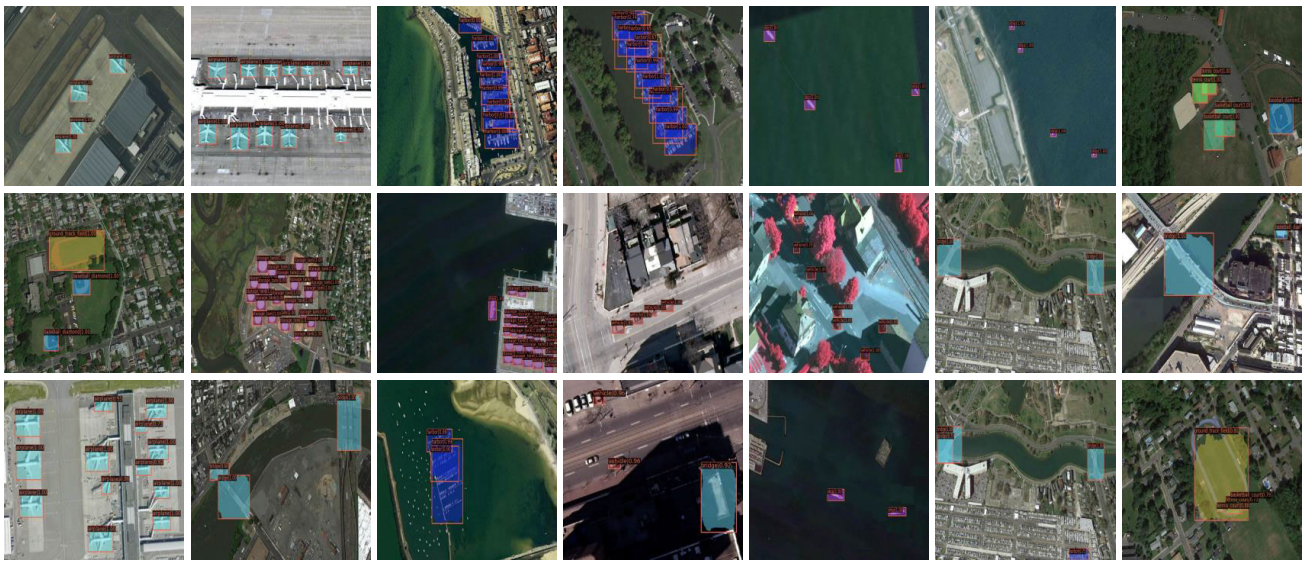


Fig. 7. Example results on the NWPU VHR-10 test split (96.61%). The first two rows correspond to success cases. The third rows indicate a few missed objects and FPs (categories: 1—airplane, 2—ship, 3—storage tank, 4—baseball diamond, 5—tennis court, 6—basketball court, 7—ground track field, 8—harbor, 9—bridge, and 10—vehicle).

that PCViT exhibits superior feature extraction capabilities. The incorporation of our multiscale backbone structure and PCM module significantly bolsters PCViT's performance. Remarkably, PCViT's performance on the DIOR dataset surpasses ViT-B by 2.07%, solidifying its position as an exceptional detector.

Subsequently, based on the PCViT model pretrained on the MillionAID dataset, we proceed to quantitatively and qualitatively evaluate the efficacy of the proposed optimization modules, namely the LGKA and FRPN, during the finetuning stage. We utilize the mAP_{50} metric for comprehensive assessment while also analyzing the distinct roles of each module.

Quantitative results from the ablation experiments are presented in Table IV, revealing that the LGKA module enhances overall accuracy by 0.28% in terms of mAP_{50}. Notably, it significantly improves large object detection, with enhancements of 2.07%, 1.33%, 1.10%, 1.09%, and 0.95% observed in chimneys, railway stations, dams, windmills, and tennis courts, respectively. Moreover, performance improvements of 0.85%, 0.78%, 0.18%, 0.17%, and 0.64% are achieved in baseball stadiums, airplanes, airports, ports, and oil storage tanks, respectively. Notably, the LGKA module also optimizes the computational complexity of the original global MHSA mechanism, contributing to enhanced network efficiency.

Furthermore, the transition from FPN to FRPN brings about a notable 0.38% enhancement in terms of mAP_{50} when applied to the DIOR dataset. This adjustment facilitates more comprehensive contextual information fusion, resulting in improved performance for objects of varying sizes. Specific gains include 2.90%, 1.60%, 1.32%, and 1.32% improvements in the accuracy of railway stations, dams, golf courses, and high-speed service areas. Similarly, improvements of 0.17%, 0.92%, and 0.80% are observed in playgrounds, tennis courts, and baseball fields, respectively.

Simultaneously incorporating both the LGKA and FRPN modules, our PCViT model demonstrates its peak performance, showcasing robust representation capabilities. The overall mAP_{50} accuracy increases by 0.69%. Notably, visualization through a heatmap in Fig. 6 illustrates the network's feature extraction process. The absence of pretrained weights in PCViT initially limits accuracy due to network depth, with a broad focus that hinders precise target localization. Integration of pretrained weights and subsequent finetuning improves region-of-interest localization. The LGKA module further refines target localization, while the combined impact of the LGKA and FRPN ensures a clear and accurate representation of target information.

2) Comparison With the State-of-the-Art: We compare the detection performance of PCViT with state-of-the-art methods, encompassing four CNN-based approaches and three ViT-based methods. Among the CNN-based methods are Faster R-CNN [10] and Mask R-CNN [11], which exhibit excellent performance in general detection tasks. Additionally, we consider Oriented R-CNN [58] and ReDet [19], specifically tailored for the characteristics of RSIs, offering robust detection capabilities. As for ViT-based methods, we employ ViTDet [25] as the baseline, utilizing ViT-B [42], ViTAE-B [30], and Swin-B [27] as backbones. These algorithms represent the latest advancements in RSOD in recent years.

Some examples of detection results in each class obtained by the PCViT network on the DIOR dataset and the NWPU VHR-10 dataset are provided in Figs. 7 and 8, respectively. As we can see, the PCViT network has detected and located most of the objects successfully despite the large variations in the directions and sizes on the two datasets. The failure results are present in Figs. 7 and 8 (bottom row). The PCViT network demonstrates poor performance in the classes of the basketball court and tennis court with false detection. Due to lacking instance-level supervision, the PCViT network sometimes misjudges the categories of the basketball court and tennis court that usually appear together in the same image.

To further demonstrate the effectiveness of our proposed network, quantitative results of our approach and state-of-the-art detectors on the DIOR dataset and the NWPU VHR-10

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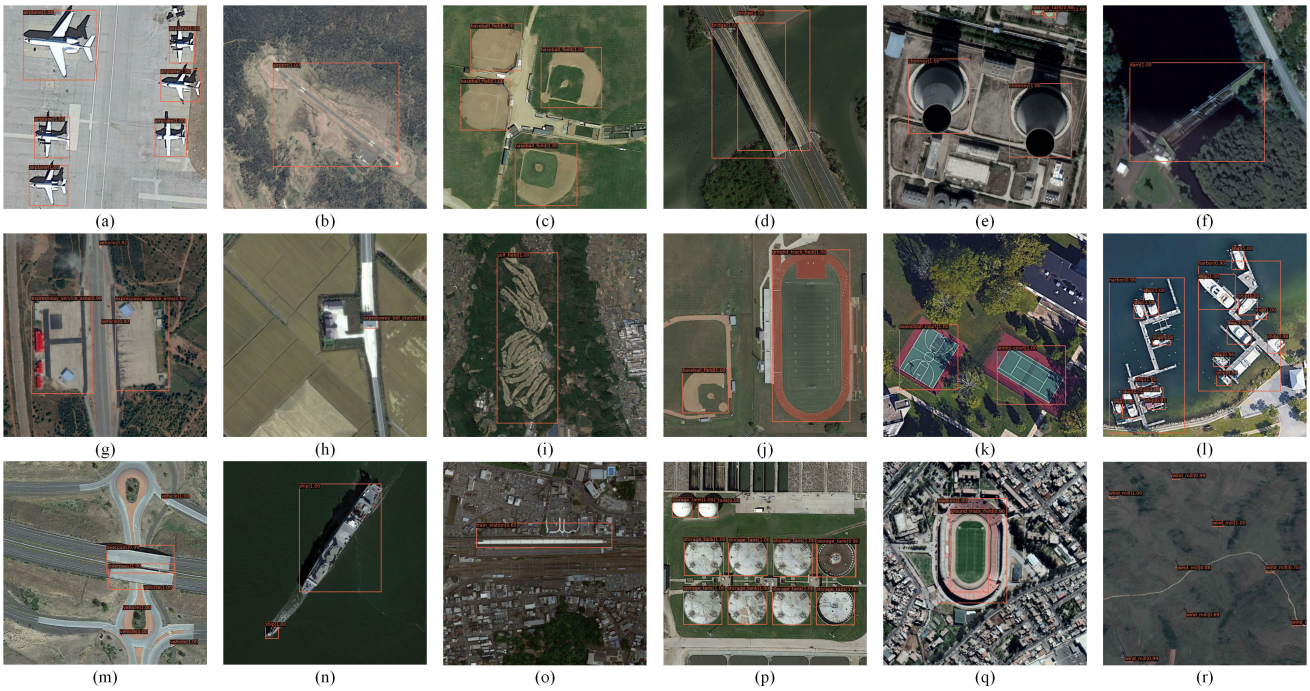


Fig. 8. Some representative detection results of PCViT on the DIOR dataset. (a) Airplane. (b) Airport. (c) Baseball field. (d) Bridge. (e) Chimney and storage tank. (f) Dam. (g) Expressway service area and vehicle. (h) Expressway toll station. (i) Golf field. (j) Ground track field and baseball field. (k) Basketball court and tennis court. (l) Harbor and ship. (m) Overpass and vehicle. (n) Ship. (o) Train station. (p) Storage tank. (q) Stadium and ground track field. (r) Wind mill.

TABLE V

COMPARISON OF THE PERFORMANCE OF OTHER METHODS ON THE NWPU-VHR-10 DATASET

No.	Faster R-CNN [10]	Mask R-CNN [11]	Oriented R-CNN [18]	ReDet [19]	ViT-B [42]	ViTAE-B [30]	Swin-B [27]	PCViT
1	99.94	99.98	**99.99**	99.92	99.98	99.98	99.98	99.94
2	89.68	90.85	89.46	**93.52**	91.01	87.78	89.63	91.42
3	95.55	95.15	94.22	**98.91**	96.97	97.82	94.99	98.68
4	94.78	96.39	96.53	94.32	96.37	96.35	**96.82**	96.77
5	91.01	89.07	91.89	95.48	95.71	94.54	**96.98**	96.26
6	99.40	95.95	99.50	95.50	99.88	97.46	99.88	**100**
7	**100**	**100**	**100**	**100**	99.72	**100**	**100**	**100**
8	92.26	90.32	**92.89**	88.07	90.13	86.91	92.29	91.78
9	63.87	79.65	72.65	70.86	71.72	89.48	91.37	**95.22**
10	91.39	94.10	95.80	**96.05**	94.30	95.98	94.56	96.04
mAP_{50}	91.78	93.15	93.29	93.26	93.58	94.63	95.65	**96.61**
mAP_s	49.04	65.61	76.69	47.26	81.35	81.25	65.68	**98.92**
mAP_m	94.86	94.28	95.49	95.88	95.92	95.29	96.00	**96.85**
mAP_l	93.48	95.97	94.58	92.54	94.02	95.98	97.85	**98.18**

dataset are presented in Tables V and VI and Figs. 9 and 10, respectively. The mAP_{50}, mAP_s, mAP_m, and mAP_l scores for 20 classes of objects intuitively illustrate the detection performance of each algorithm.

In comparison to CNN-based networks, the PCViT network demonstrates superior mAP on the NWPU VHR-10 dataset, outperforming classic Faster R-CNN by 5.83%, Oriented R-CNN by 3.46%, and ReDet by 3.32%. On the DIOR dataset, PCViT achieves mAP improvements of 8.37%, 6.77%, and 4.07% compared to Faster R-CNN, Oriented R-CNN, and ReDet, respectively. This may be attributed to the stronger global feature extraction capability of the ViT network over the CNN network.

Furthermore, a comparative analysis was conducted between our proposed PCViT and other ViT backbones. On the NWPU VHR-10 dataset, PCViT exhibits an mAP increase of 3.03%, 1.98%, and 0.96% over ViT-B, ViTAE-B, and Swin-B, respectively. Similarly, on the DIOR dataset, PCViT demonstrates noteworthy performance, surpassing ViT-B by 3.24%, ViTAE-B by 0.96%, and Swin-B by 3.80%. Besides, PCViT demonstrates superior performance in terms of mAP_{50}, mAP_s, mAP_m, and mAP_l across both datasets. These results underscore the remarkable detection performance achieved in RSOD tasks, benefiting from PCViT's robust capabilities in both global and local feature extraction.

Examining PRC curves in Fig. 11, PCViT's capacity for global and local extraction becomes more apparent across various DIOR categories. It excels in larger targets like bridges, railway stations, playgrounds, and airports, reflected by fuller PRC curves and higher AP in these categories. In the detection of smaller targets like cars and ships, PCViT maintains competitive accuracy levels, effectively demonstrating its robustness.

The robustness and effectiveness of the PCViT network are conclusively evidenced through comprehensive experiments. The method consistently demonstrates superior performance, particularly for large-scale objects such as bridges, harbors, and train stations, highlighting PCViT's prowess in global

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TABLE VI
COMPARISON OF THE PERFORMANCE OF OTHER METHODS ON THE DIOR DATASET

No.	Faster R-CNN [10]	Mask R-CNN [11]	Oriented R-CNN [18]	ReDet [19]	ViT-B [42]	ViTAE-B [30]	Swin-B [27]	PCViT
1	90.98	-	92.88	91.08	**93.03**	**93.03**	92.05	92.90
2	70.22	-	70.85	78.56	83.54	80.53	78.44	**84.49**
3	92.80	-	90.94	91.92	92.75	92.00	91.99	**93.77**
4	81.16	-	81.89	86.44	83.03	84.86	83.70	**87.59**
5	43.02	-	46.85	51.87	51.02	59.01	53.45	**59.63**
6	89.74	-	89.85	**91.88**	90.96	91.65	89.89	91.78
7	49.67	-	59.47	62.97	69.85	74.28	65.84	**74.47**
8	66.80	-	67.12	68.09	68.91	**72.61**	68.81	70.28
9	57.88	-	59.15	**63.04**	58.89	60.94	59.83	62.92
10	77.89	-	76.89	78.22	82.90	82.92	82.30	**84.56**
11	76.27	-	75.00	77.68	80.56	82.57	83.38	**84.86**
12	52.74	-	49.34	58.59	57.50	60.93	57.41	**61.40**
13	63.99	-	63.78	67.58	66.61	68.74	67.19	**70.27**
14	79.35	-	79.48	**80.39**	79.36	79.37	80.38	80.37
15	92.70	-	89.52	91.29	95.23	**96.58**	89.75	96.17
16	80.54	-	81.60	80.72	83.87	85.01	82.59	**85.74**
17	89.77	-	87.03	88.05	89.83	91.90	88.05	**92.77**
18	48.42	-	55.39	64.69	67.89	**74.83**	62.27	73.35
19	73.13	-	74.22	74.55	74.55	77.92	74.56	**79.56**
20	75.27	-	78.37	76.03	**79.60**	76.20	77.06	78.11
mAP_{50}	72.62	-	73.48	76.18	77.49	79.29	76.45	**80.25**
mAP_s	18.66	-	17.85	16.84	20.57	20.94	18.37	**21.78**
mAP_m	47.42	-	46.38	51.06	51.22	54.82	52.06	**58.24**
mAP_l	84.80	-	85.96	88.45	89.57	90.22	88.73	**90.96**

[1] Note: Since the dior data set lacks the mask label, we cannot complete the evaluation of the Mask R-CNN algorithm on the dior data set. Here, "-" is used to replace the experimental results.

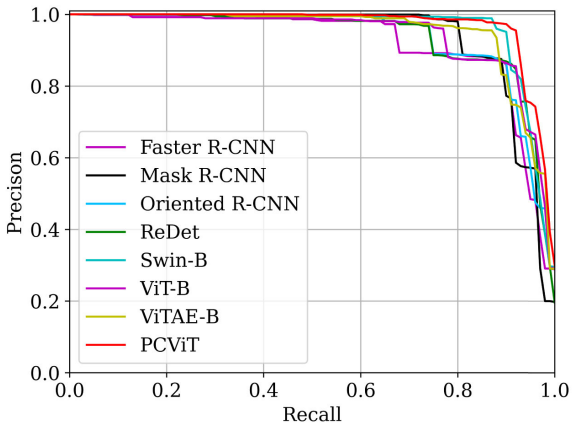


Fig. 9. PRCs of the proposed PCViT network and other state-of-the-art approaches on the NWPU VHR-10 dataset.

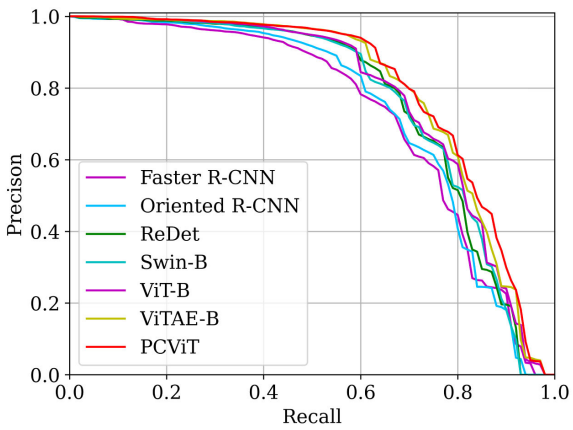


Fig. 10. PRCs of the proposed PCViT network and other state-of-the-art approaches on the DIOR dataset.

modeling and effective long-distance information capture. Further affirmation is provided through visualization of detection outcomes in Fig. 8 on the DIOR dataset.

The computational efficiency of algorithms plays a pivotal role in remote-sensing applications. To address this concern, we meticulously evaluated the computational cost associated with our proposed method. Specifically, tests were conducted using input images sized at 512 × 512 pixels to assess the computational demands of various algorithms. Our analysis primarily focused on two critical metrics: model parameters and computational cost, detailed in Table VII.

In Table VII, the comparative analysis showcases that ViT-based object detection systems tend to exhibit higher model parameter counts and computational overhead compared to their CNN-based counterparts. However, when evaluated within the context of the same ViT architecture, our proposed PCViT demonstrates notably superior performance. Despite a modest increase in parameter count, PCViT substantially improves detection performance in contrast to other ViT-based detectors.

TABLE VII
COMPUTATIONAL COST OF DIFFERENT METHODS

Method	Params (M)	Flops (G)
Faster R-CNN [10]	41.17	69.29
Mask R-CNN [11]	43.79	114.95
Oriented R-CNN [18]	41.14	63.3
ReDet [19]	**31.64**	**40.98**
Swin-B [27]	68.75	140.05
ViT-B [42]	115.80	339.44
VITAE-B [30]	118.57	335.03
PCViT	112.36	349.07

These visual results yield two conclusions.

1) The global modeling capability of the Transformer in PCViT improves the detection performance of large-scale objects: for example, in the detection results of the proposed PCViT network, categories such as "airport," "ridge," "ground track field," and "train station" are

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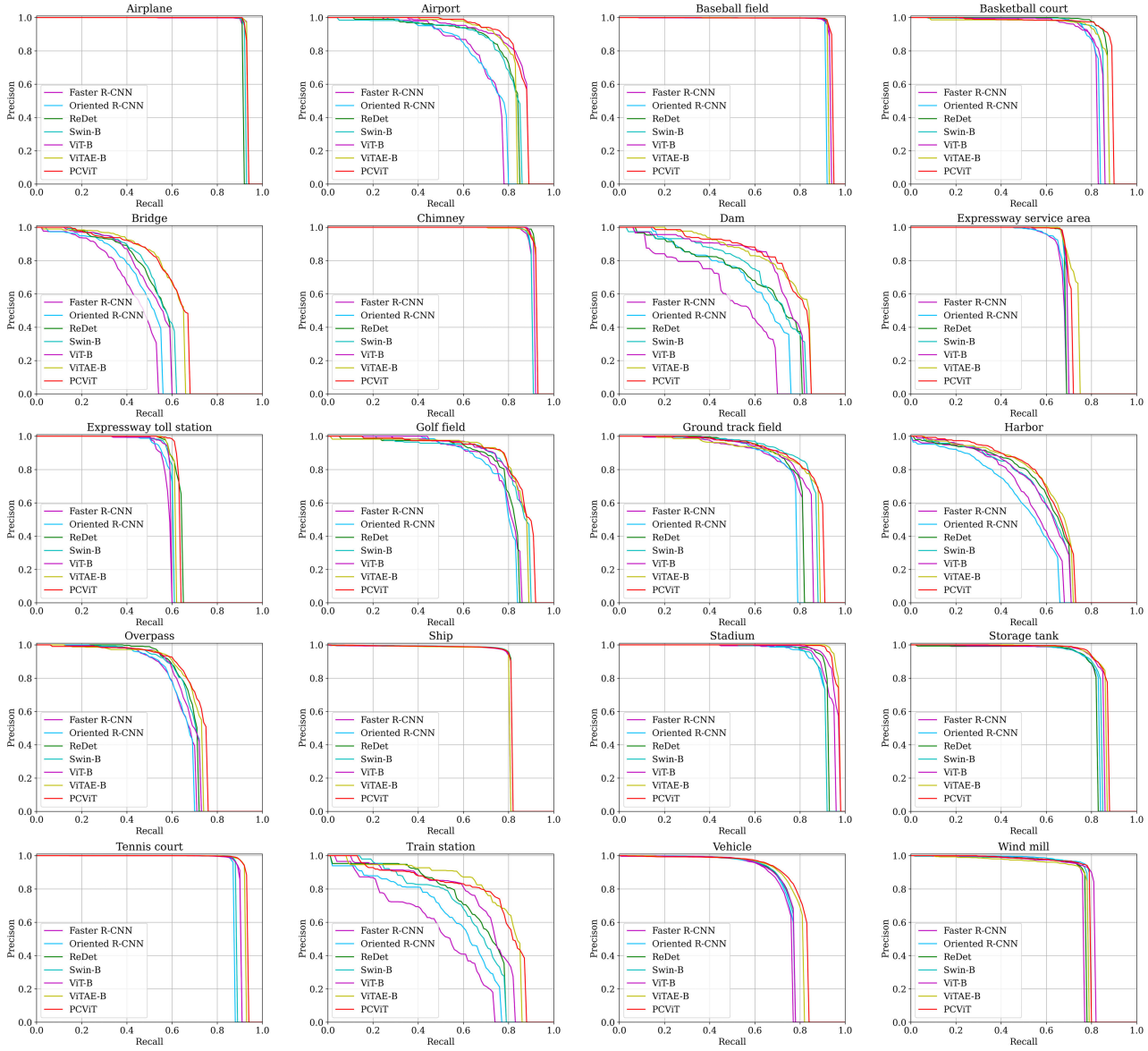


Fig. 11. Class-wise PRCs of the proposed PCViT network and other state-of-the-art approaches on the DIOR dataset.

similar. Compared with the CNN network, mAP_{50} has been greatly improved.

2) At the same time, the specially designed PCM in PCViT and the convolution bias introduced by LGKA can well supplement the local information missing from the Transformer. From the detection results of the PCViT network, the "vehicle" on the "bridge" and the "ship" near the "harbor" can be detected well in large scenes.

V. Conclusion

In this article, a pyramidal detector hybrid of the CNN and Transformer is proposed for RSOD. First of all, we designed the pyramid backbone called PCViT, which retains multi-scale information well. The Transformer block exerts the global modeling ability of MHSA. At the same time, the parallel CNN module supplements local information well. Then, we trained the PCViT to a better network weight on MillionAID through the MPP method and then fine-tuned on the downstream RSOD task. During the finetuning process, we proposed the LGKA, which better characterizes the relationship between tokens and strengthens the attention to the most important information. Finally, we optimize the original FPN structure, strengthen the interaction of context information through reflowing, and enhance the detection performance of the network for multiscale targets. Our PCViT is demonstrated to far outperform existing SOTA algorithms on two public benchmarks. However, our algorithm is still a large model. In the future, we will optimize the weight and speed of the model and pay attention to the rotation of the target in RSIs.

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